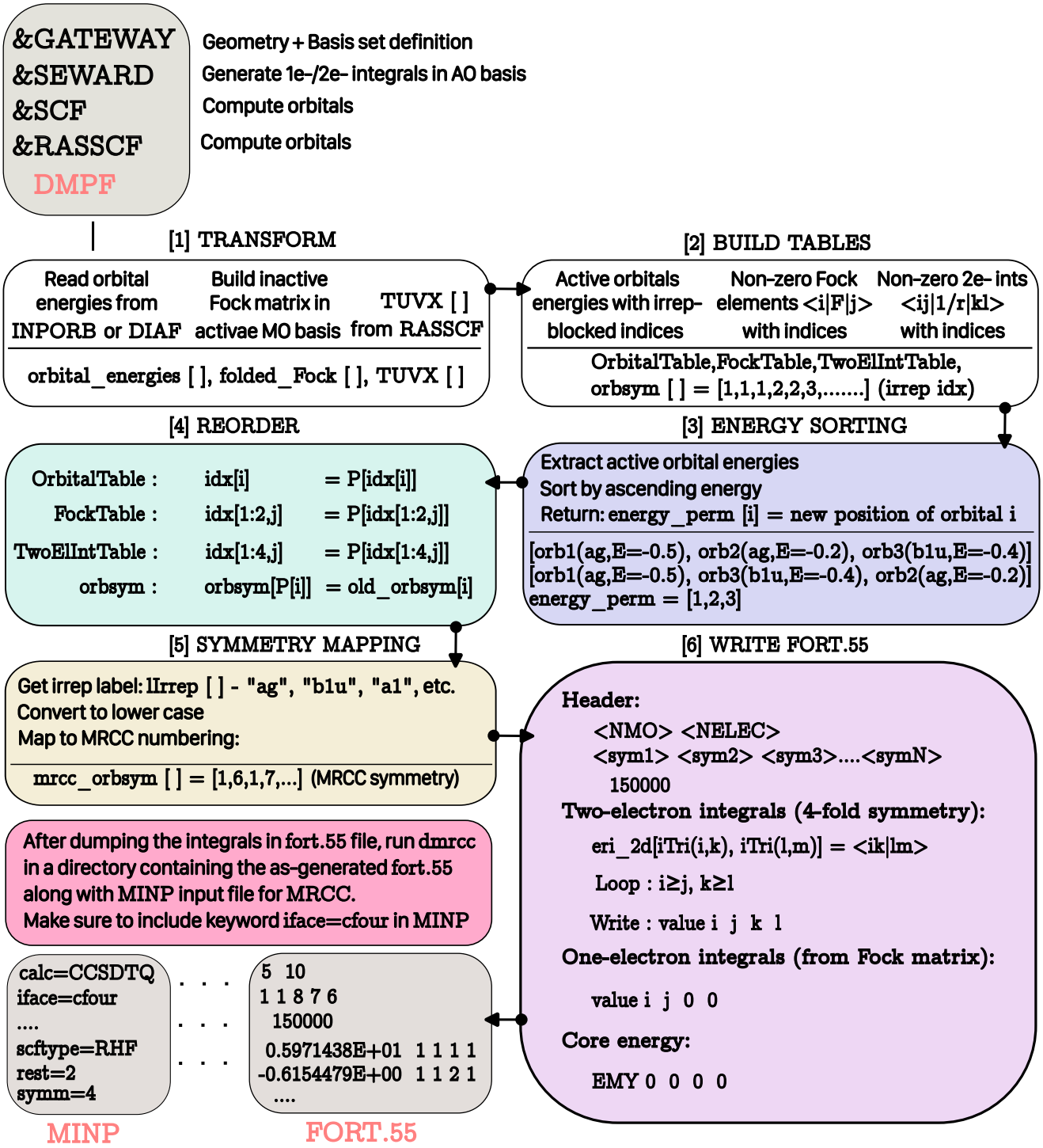

&GATEWAY
&SEWARD
&SCF
&RASSCF
DMPF
Geometry + Basis set definition
Generate 1e-/2e- integrals in AO basis
Compute orbitals
Compute orbitals
[1] TRANSFORM
Read orbital energies from INPORB or DIAF
Build inactive Fock matrix in activae MO basis
TUVX [ ] from RASSCF
orbital_energies [ ], folded_Fock [ ], TUVX [ ]
[2] BUILD TABLES
Active orbitals energies with irrep-blocked indices
Non-zero Fock elements <i|F|j> with indices
Non-zero 2e- ints <ij|1/r|kl> with indices
OrbitalTable,FockTable,TwoElIntTable,
orbsym [ ] = [1,1,1,2,2,3,.....] (irrep idx)
[3] ENERGY SORTING
Extract active orbital energies
Sort by ascending energy
Return: energy_perm [i] = new position of orbital i
[orb1(ag,E=-0.5), orb2(ag,E=-0.2), orb3(b1u,E=-0.4)]
[orb1(ag,E=-0.5), orb3(b1u,E=-0.4), orb2(ag,E=-0.2)]
energy_perm = [1,2,3]
[4] REORDER
OrbitalTable : idx[i] = P[idx[i]]
FockTable : idx[1:2,j] = P[idx[1:2,j]]
TwoElIntTable : idx[1:4,j] = P[idx[1:4,j]]
orbsym : orbsym[P[i]] = old_orbsym[i]
[5] SYMMETRY MAPPING
Get irrep label: lIrrep [ ] - "ag", "b1u", "a1", etc.
Convert to lower case
Map to MRCC numbering:
mrcc_orbsym [ ] = [1,6,1,7,...] (MRCC symmetry)
[6] WRITE FORT.55
Header:
<NMO> <NELEC>
<sym1> <sym2> <sym3>....<symN>
150000
Two-electron integrals (4-fold symmetry):
eri_2d[iTri(i,k), iTri(l,m)] = <ik|lm>
Loop : i≥j, k≥l
Write : value i j k l
One-electron integrals (from Fock matrix):
value i j 0 0
Core energy:
EMY 0 0 0 0
After dumping the integrals in fort.55 file, run dmrcc in a directory containing the as-generated fort.55 along with MINP input file for MRCC.
Make sure to include keyword iface=cfour in MINP
calc=CCSDTQ
iface=cfour
....
scftype=RHF
rest=2
symm=4
MINP
5 10
1 1 8 7 6
150000
0.5971438E+01 1 1 1 1
-0.6154479E+00 1 1 2 1
....
FORT.55